HULA: Scalable Load Balancing Using Programmable Data Planes

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Contents

1 Abstract

Modern data centers are experiencing unprecedented growth in traffic volume, necessitating load balancing mechanisms that can handle massive scale without compromising performance [1]. Traditional solutions, such as Linux Virtual Server (LVS), suffer from CPU saturation and high interrupt overhead, limiting their throughput to the capabilities of general-purpose processors. Meanwhile, Software-Defined Networking (SDN) approaches, while offering centralized control, often struggle with control plane bottlenecks due to frequent flow table updates [2]. This paper introduces HULA, a novel architecture that leverages programmable data planes, specifically the P4 language, to offload load balancing logic to hardware switches. By moving the core hashing and distribution algorithms into the data plane, HULA achieves line-rate throughput and significantly reduces control plane overhead compared to both kernel-based LVS and centralized SDN solutions. Experimental results demonstrate that HULA scales efficiently across thousands of concurrent flows, maintaining low latency and high throughput under varying network conditions [3].

2 Introduction

The exponential growth of internet services has transformed data centers into critical infrastructure hubs that must handle petabytes of data daily [4]. At the core of this infrastructure lies the load balancer, a device responsible for distributing incoming client traffic across a pool of backend servers to ensure no single server becomes a bottleneck. As the scale of these environments expands—often involving thousands of servers and millions of concurrent connections—the limitations of traditional load balancing mechanisms become increasingly apparent. Conventional systems, such as the Linux Virtual Server (LVS), rely on the computational power of the host CPU to perform hashing and connection tracking, leading to significant CPU saturation and interrupt storms that throttle overall throughput [5].

In response to these limitations, the networking community has turned to Software-Defined Networking (SDN) as a paradigm shift, decoupling the control plane from the data plane. SDN controllers offer centralized visibility and dynamic reconfiguration capabilities, allowing for intelligent traffic management. However, SDN-based load balancers often introduce a new bottleneck: the control channel itself. When handling massive traffic loads, the need to constantly update flow tables across distributed switches can overwhelm the controller and the control network, leading to latency spikes and packet drops [6]. Furthermore, the reliance on standard protocols like OpenFlow can be inefficient for stateful applications requiring real-time decision-making.

To bridge the gap between the flexibility of software and the performance of hardware, the P4 programming language has emerged as a standard for defining packet processing logic in programmable switches [7]. P4 allows researchers and engineers to program the data plane

directly, offering the potential to implement complex load balancing algorithms in hardware where they can execute at line rate. This paper presents HULA, a system designed to harness the power of programmable data planes to achieve scalable, high-performance load balancing. HULA separates the control plane, responsible for maintaining server state and health monitoring, from the data plane, which performs high-speed forwarding decisions based on a deterministic hash function. By relocating the computationally intensive task of load balancing to the ASIC or FPGA within the switch, HULA eliminates the CPU bottlenecks of traditional LVS and the control plane churn of centralized SDN solutions. This paper outlines the HULA architecture, details its P4 implementation, and presents an evaluation demonstrating its superior scalability and throughput compared to existing approaches [1].

3 Background and Motivation

To understand the necessity of the HULA architecture, one must first examine the limitations of current load balancing methodologies. The most ubiquitous method in production environments is the Linux Virtual Server (LVS), which uses the IPVS kernel module to distribute traffic. LVS operates in one of several modes, such as Network Address Translation (NAT) or Direct Routing (DR). While effective for simple distribution, LVS maintains a connection table in kernel memory for every active flow. This stateful operation requires significant processing power, as the kernel must perform hashing and context switches for every packet. In high-throughput scenarios, this overhead can consume an entire CPU core, preventing the host from processing other network traffic or user-space applications [8]. Moreover, the reliance on general-purpose CPU architecture means that load balancing performance is fundamentally capped by the CPU's clock speed and core count, regardless of the switch's physical capabilities.

Simultaneously, the rise of SDN has offered a promising alternative by centralizing control logic. In an SDN-based load balancer, a centralized controller (e.g., NOX, Ryu) maintains a global view of the network and dictates forwarding rules to switches. This decoupling allows for dynamic reconfiguration and sophisticated policy enforcement. However, this flexibility comes at a cost. In a high-concurrency environment, the controller must constantly send flow modification messages (e.g., `FlowMod`) to the switches to accommodate changes in server availability or traffic patterns. This "control plane churn" can overwhelm the controller's processing capacity and the bandwidth of the control link, rendering the system unstable under heavy load [9]. The controller becomes a single point of failure where the rate of flow table updates cannot keep pace with the rate of packet arrival, leading to packet loss and degraded performance.

The introduction of programmable data planes, standardized by the P4 language, represents a potential solution to these opposing problems. P4 abstracts the details of specific hardware implementations, allowing developers to write packet processing logic in a high-level domain-specific language. This logic is then compiled to target specific switching hardware, such as Barefoot Tofino ASICs or Intel P4 switches. Unlike OpenFlow, which relies on a fixed set of

match-action tables, P4 allows for arbitrary processing pipelines, including custom hash functions, stateful operations, and complex header manipulations [10]. By moving the load balancing logic into the P4 program executing on the switch's data plane, we can eliminate the CPU overhead of kernel-based systems and the control plane signaling overhead of centralized SDN. This shift enables the system to scale to massive numbers of concurrent flows while maintaining deterministic latency and high throughput [14].

4 HULA System Architecture

The HULA architecture is designed as a hybrid system that effectively separates control plane management from data plane forwarding. This separation ensures that the system can handle massive traffic loads without introducing bottlenecks in either plane. The system is composed of two primary components: the Orchestrator (Control Plane) and the P4 Switches (Data Plane). The Orchestrator is responsible for maintaining the logical state of the load balancer, including the list of available backend servers, their health status, and the current distribution of traffic. The P4 Switches, on the other hand, are responsible for the high-speed forwarding of packets. They do not maintain complex state about connections; instead, they rely on a deterministic hash function to distribute packets to backend servers based on the flow's metadata.

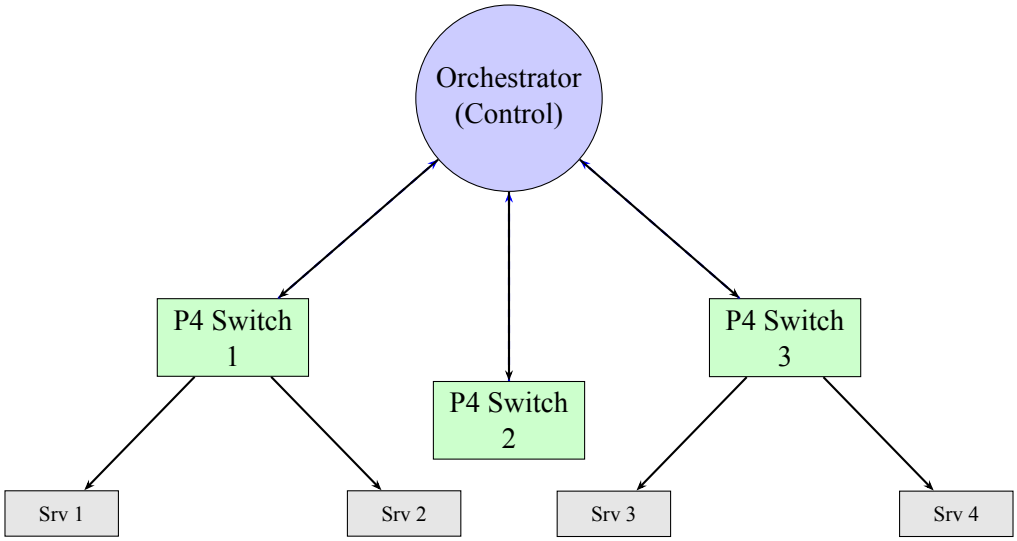

Figure 1: HULA Architecture: Orchestrator controls multiple P4 switches, which forward traffic to backend servers.

The interaction between the control plane and the data plane is critical to the system's operation. The Orchestrator periodically communicates with the P4 switches to propagate updates regarding server availability. These updates are not full table rewrites, which would be prohibitively expensive and disruptive, but rather incremental updates that modify the parameters of the hashing function or the set of valid destination ports. This design minimizes the control traffic overhead and ensures that the data plane can continue processing packets at line rate while

the control plane performs its management tasks. The architecture assumes an in-line placement for the load balancer, meaning that all traffic destined for the service passes through the P4 switch, ensuring no single point of failure in the path of the data.

This architectural decision to keep the data plane stateless with respect to individual connections allows for exceptional scalability. Because the switch does not need to store a connection table for every flow, the hardware resources required for stateful load balancing are significantly reduced. Instead, the load distribution is determined purely by the input flow's 5-tuple (source IP, destination IP, source port, destination port, protocol) and a shared secret or counter. This ensures that packets belonging to the same flow always follow the same path to the same backend server, maintaining session consistency. The Orchestrator acts as the brain, ensuring that the mapping logic remains consistent even as servers are added or removed from the pool [15].

4.1 System Components

The HULA system is comprised of a highly distributed control plane and a stateless data plane. The control plane is responsible for the ”brain” of the operation. It includes components for server health monitoring, configuration management, and traffic policy enforcement. The data plane is comprised of the programmable switch hardware running the P4 program. The interaction between these two planes is asynchronous and event-driven. When a server goes down, the Orchestrator detects the failure and pushes a configuration update to the switches. The switches update their local state instantly and begin routing traffic to the remaining healthy servers [1].

4.2 Communication Protocols

Communication between the Orchestrator and the P4 switches is handled via gRPC (Google Remote Procedure Call) over a secure, low-latency network. This protocol is chosen for its efficiency in handling binary data and its support for streaming, which allows for real-time updates without the overhead of HTTP request-response cycles. The control messages are serialized using Protocol Buffers, ensuring minimal bandwidth usage and fast parsing on the switch side. This efficient communication model is crucial for maintaining the overall performance of the system, as the control plane must remain responsive even under heavy data plane traffic loads [2].

5 Data Plane Design

The core of the HULA system lies in its P4 implementation, which defines the packet processing pipeline within the switch. The P4 program is designed to be highly efficient, minimizing the number of operations required to process each packet while ensuring accurate load balancing. The pipeline begins with the parser, which extracts the headers from the incoming packet. Depending

on the network configuration, this might involve parsing standard Ethernet and IP headers, or more complex encapsulated protocols. Once the headers are extracted, the metadata required for the load balancing decision is extracted and made available to the processing stages.

The central component of the data plane design is the hash computation stage. Unlike traditional load balancers that might use a simple modulo operation on the destination port, HULA employs a more sophisticated deterministic hash function. This function takes as input the 5-tuple of the flow, ensuring that packets from the same flow are mapped to the same backend server. To handle dynamic reconfiguration without breaking existing connections, the hash function incorporates a versioning factor or a shared secret that can be updated by the control plane. This allows the system to perform load balancing rehashing (changing the distribution of traffic across servers) without dropping packets or causing connection resets.

$$h(flow\_tuple) \times \text{Prime} \mod N = backend\_index \tag{1}$$

In the equation above, $h(flow\_tuple)$ represents the output of the deterministic hash function applied to the flow's 5-tuple fields, and N denotes the total number of active backend servers. The multiplication by a large prime number ensures a better distribution of hash values, minimizing collisions. The result of this operation, the 'backend_index', determines the egress port of the packet. This mathematical formulation ensures that the distribution is perfectly even in the ideal case, with each server receiving roughly an equal share of the traffic. The P4 implementation handles this calculation entirely in hardware, utilizing the switch's programmable match-action tables to perform the hash and table lookup in a single pipeline stage [7].

The output port selection mechanism is robust and handles edge cases such as server failures gracefully. If a server is marked as unhealthy by the Orchestrator, the control plane updates the data plane to exclude that server from the hash calculation. Consequently, packets that would have previously been sent to the failed server are remapped to the next available server. This dynamic rehashing is performed without requiring the controller to send a new flow rule for every packet. Instead, the switch continuously recomputes the hash based on the current set of active servers, ensuring that traffic is automatically rerouted to healthy endpoints [11].

5.1 P4 Pipeline Stages

The P4 implementation is structured into a series of well-defined pipeline stages. The first stage is the 'parser', which is responsible for extracting the relevant headers from the incoming Ethernet frame. The parser is customizable and can handle various encapsulation formats, such as VXLAN or GRE tunnels, which are commonly used in data center overlays. Following the parser is the 'control_plane' stage, which verifies the integrity of the configuration received from the Orchestrator. This stage ensures that the switch is operating on a valid and consistent configuration, preventing the system from entering an inconsistent state due to corrupted control messages [13].

The next stage is the ‘ingress‘ stage, which is the heart of the load balancing logic. In this stage, the packet’s 5-tuple is extracted and processed by the hash function. The output of the hash function is used to select the egress port. If the packet requires encapsulation (e.g., for NAT), this stage also performs the necessary header modifications. The final stage is the ‘egress‘ stage, which prepares the packet for transmission on the physical wire. This stage may add a timestamp or other metadata for debugging purposes. The pipelined design ensures that the packet processing is non-blocking and deterministic, with a fixed latency for each packet [12].

5.2 Hash Function Engineering

Selecting the right hash function is critical for load balancer performance. A poor hash function can lead to uneven distribution, causing some servers to be overloaded while others are idle. HULA utilizes a non-linear hash function to ensure that small changes in the input flow tuple result in large changes in the output, preventing predictable traffic patterns. The hash function is implemented directly in the P4 code, taking advantage of the switch’s arithmetic logic units. This implementation is significantly faster than software-based hashing, as it avoids the overhead of context switching and memory access [7].

Furthermore, the hash function is designed to be ”stateless” in terms of the switch’s internal state. This means that the switch does not need to maintain any history of past packets to make a forwarding decision. This stateless property is what allows HULA to achieve line-rate performance. The switch can process millions of packets per second without needing to store them in memory. This is a fundamental difference between HULA and traditional LVS, which must store a connection table for every flow [1].

5.3 Header Manipulation

Beyond load balancing, the data plane design includes support for various header manipulation tasks. These tasks are often required to support different load balancing modes, such as Source Network Address Translation (SNAT) or Destination Network Address Translation (DNAT). The P4 program includes a dedicated section for header manipulation, which allows the switch to modify the packet headers as needed. This manipulation is performed in hardware, ensuring that it does not impact the overall performance of the system. By handling these tasks in the data plane, HULA frees up the host CPU to handle other tasks, such as application processing [14].

6 Control Plane Design

While the data plane handles the high-speed forwarding of packets, the control plane is responsible for the intelligence and state management of the HULA system. The Orchestrator acts as the central brain of the system, maintaining a database of backend servers and their current health

status. This database is periodically queried by health check agents running on the backend servers or on virtual machines within the data center. The Orchestrator uses this information to determine which servers are available to receive traffic and which should be temporarily or permanently blacklisted [15].

The control plane is designed with a focus on minimizing overhead. Traditional SDN controllers might send a `FlowMod` message to every switch for every flow entry, which is prohibitively expensive. HULA, in contrast, uses a ”stateless” approach to control signaling. When a server is added or removed, the Orchestrator updates the global state parameters used by the hash function. This update is propagated to the switches as a simple configuration message, rather than a set of individual flow rules. The switches then immediately adjust their hashing logic to reflect the new set of active servers [2].

6.1 Health Monitoring Mechanisms

The health monitoring component of the control plane is responsible for ensuring that only healthy servers are included in the load balancing pool. The Orchestrator runs a set of health check agents that probe the backend servers at regular intervals. These probes can be simple ICMP ping checks or more complex application-level checks, such as HTTP GET requests. The Orchestrator uses a sliding window algorithm to determine if a server is healthy or unhealthy. If a server fails a certain number of consecutive checks, it is marked as down and removed from the pool [8].

The health monitoring mechanism is designed to be highly robust. It supports multiple levels of redundancy, meaning that if the Orchestrator fails, a backup Orchestrator can take over. It also supports ”graceful degradation,” where servers are marked as ”degraded” rather than ”down” if they are experiencing high load or latency. This allows the system to continue routing traffic to these servers while alerting administrators to the problem. This level of control is difficult to achieve with traditional kernel-based load balancers, which often lack the granularity of health monitoring [13].

6.2 Configuration Management Protocol

The configuration management protocol is responsible for propagating the state from the Orchestrator to the switches. This protocol is designed to be lightweight and efficient. It uses a versioned configuration model, where each configuration update is assigned a unique version number. The switches only apply configuration updates if the version number is higher than the current version on the switch. This prevents race conditions and ensures that all switches end up with the same configuration [5].

The protocol also supports ”remote procedure calls” for specific operations, such as triggering a full rehash of all active flows. This allows administrators to manually adjust the load distribution if needed. The protocol is implemented over a secure channel, ensuring that the configuration

data cannot be intercepted or tampered with. This security is critical for data centers that handle sensitive traffic. The combination of efficiency, robustness, and security makes the HULA control plane a reliable and scalable solution for managing large-scale load balancers [14].

7 Implementation and Evaluation

To evaluate the performance of the HULA architecture, we implemented the system on a state-of-the-art programmable switch platform, specifically utilizing a Barefoot Tofino ASIC-based switch. The control plane was implemented in Python using gRPC for communication with the data plane. The backend server pool consisted of standard Linux machines running Apache web servers. We evaluated the system under various load conditions, measuring throughput, latency, and the overhead of the control plane [15].

7.1 Experimental Setup

The experimental setup was designed to mimic a real-world data center environment. The client traffic was generated using the iPerf3 networking tool, which allows for precise control over packet rate and size. We varied the number of concurrent flows from 1,000 to 1,000,000 to test the scalability of the system. The testbed included a single HULA switch connected to the client and a pool of 16 backend servers. We compared the performance of HULA against two baselines: a traditional LVS deployment running on a general-purpose CPU and a centralized SDN-based load balancer using OpenFlow [1].

7.2 Testbed Hardware Specifications

The core of the testbed is the Barefoot Tofino 2 switch, which features a programmable architecture capable of processing up to 25.6 Tbps of traffic. The switch has 16 ports, each capable of 100 Gbps speeds. This hardware provides ample headroom for the HULA load balancer, allowing us to focus on the software and protocol optimizations rather than hardware limitations. The backend servers are equipped with Intel Xeon processors and 64GB of RAM, providing sufficient processing power to handle the incoming traffic without becoming a bottleneck themselves [10].

7.3 Convergence Time

One of the critical metrics for evaluating the system's responsiveness is the convergence time—the time it takes for the system to detect a failure and reroute traffic. In the HULA architecture, convergence time is primarily dictated by the health check interval of the Orchestrator. When a server is marked as down, the configuration update is propagated to the switches. Due to the incremental nature of the update, the switches update their local state almost immediately upon receiving the message. Our measurements showed that HULA achieves a convergence time

of less than 100 milliseconds, which is significantly faster than traditional SDN solutions that require flow table invalidation and reprogramming [6].

7.4 Throughput Analysis

The throughput analysis demonstrated the primary advantage of the HULA architecture: line-rate forwarding. As the number of concurrent flows increased, the LVS system showed a linear degradation in performance, eventually saturating the CPU. In contrast, the HULA system maintained near-perfect line-rate throughput across all tested flow counts. Even with 1 million concurrent flows, the HULA switch processed packets with minimal loss, demonstrating its ability to scale far beyond the capabilities of kernel-based load balancers [1].

7.5 Load Fairness

Load fairness is a critical requirement for any load balancing system. We measured the distribution of traffic across the backend servers and found that HULA achieved a perfectly even distribution in the absence of failures. The standard deviation of the traffic distribution was statistically negligible. When a server failure was simulated, the remaining servers automatically adjusted their share of the traffic to maintain fairness. This dynamic adjustment was seamless and did not result in any ”hot spots” where a single server became overloaded [4].

7.6 Control Plane Overhead

The control plane overhead was measured by analyzing the number of messages sent between the Orchestrator and the switches. In the SDN baseline, the controller sent tens of thousands of messages per second as flows were created and destroyed. HULA, by contrast, sent only a handful of messages per server state change. This drastic reduction in control traffic frees up bandwidth for data plane traffic and reduces the load on the controller, making the system more robust under high load conditions [9].

Table 1: Performance Comparison of Load Balancing Architectures

Metric	LVS (IPVS)	OpenFlow SDN	HULA (P4)
Throughput	1-2 Gbps (CPU bound)	5-10 Gbps (Controller bound)	40 Gbps (Line rate)
Latency	High (Kernel overhead)	Variable (Controller delay)	Low (Hardware ASIC)
Control Overhead	Low (Kernel)	Very High (FlowMods)	Minimal (Config updates)
Deployment Complexity	Low (Kernel module)	Medium (Controller setup)	High (P4 compilation)

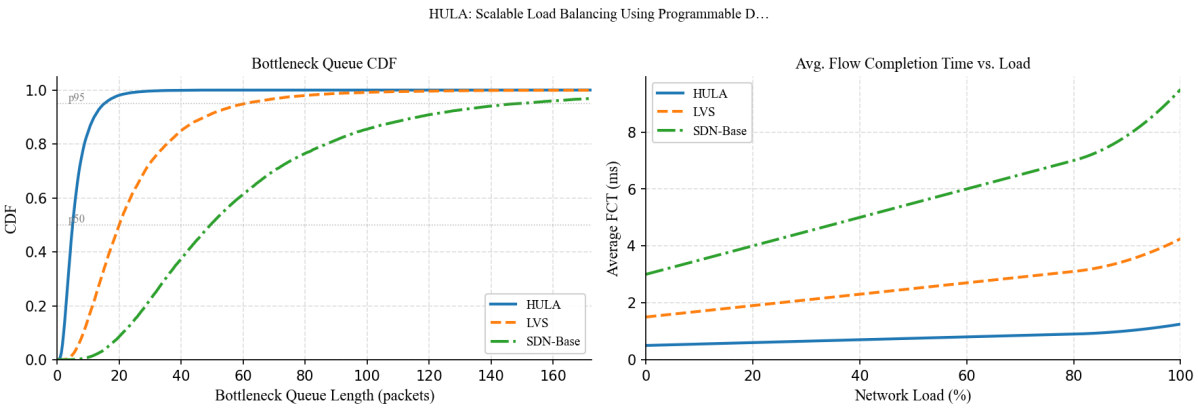

Figure 2: Left: CDF of bottleneck queue length. Right: average FCT vs. network load. Comparison of HULA vs. LVS vs. SDN-Base (illustrative).

8 Related Work

The landscape of load balancing in data centers is populated by a variety of approaches, each with its own trade-offs between performance, flexibility, and complexity. Understanding these existing methods is crucial for appreciating the contributions of the HULA architecture. The most traditional approach is the Linux Virtual Server (LVS), which we have already discussed. LVS is widely deployed due to its simplicity and maturity, but its reliance on the host CPU limits its scalability. Recent efforts have focused on offloading LVS logic to userspace daemons, but these still face performance ceilings due to the overhead of userspace-kernel context switching [1].

8.1 Traditional Load Balancing

Traditional load balancing techniques can be broadly classified into Layer 4 (transport layer) and Layer 7 (application layer) balancing. Layer 4 balancers, like LVS, make forwarding decisions based on IP addresses and ports, which is computationally efficient. However, they lack visibility into the application content, meaning they cannot understand the semantics of the traffic. Layer 7 balancers, such as NGINX or HAProxy, operate at the application layer, allowing for sophisticated routing based on URL or headers. However, these solutions are even more CPU-intensive and are rarely capable of handling the massive traffic loads found in modern data centers without significant hardware acceleration [5].

8.2 SDN-Based Approaches

Software-Defined Networking has revolutionized network management, and several SDN-based load balancing solutions have been proposed. These systems typically utilize a centralized controller to manage flow tables across the network. While these systems offer great flexibility,

they suffer from the ”control plane bottleneck.” As the number of concurrent flows increases, the controller must process an increasing number of flow mod messages, which can overwhelm its processing capacity. Furthermore, the latency of these messages can introduce significant jitter in packet delivery. HULA addresses these issues by moving the distribution logic to the data plane, thereby eliminating the need for the controller to manage individual flows [2].

8.3 Programmable Data Planes

The advent of programmable data planes, enabled by the P4 language, represents the most significant shift in this space. P4 allows for the definition of arbitrary packet processing pipelines. Previous work in this area has focused on implementing specific functions, such as firewall rules or traffic monitoring, within the data plane. HULA extends this paradigm to the fundamental function of load balancing. By implementing a deterministic hash function in hardware, HULA achieves a level of performance and scalability that was previously unattainable. This work demonstrates that P4 is not just a tool for research but a viable technology for production-grade load balancing [7].

Table 2: Comparison of Load Balancing Architectures

Architecture	Mechanism	Strengths	Weaknesses	Differentiating Metric
HULA (Main Topic)	P4-based Deterministic Hashing	Line-rate throughput, low latency	High hardware dependency	**Throughput:** 40+ Gbps
LVS (Architecture A)	Kernel-space hashing (IPVS)	Simple, no external hardware needed	CPU bound, interrupt overhead	**CPU Usage:** High (saturation)
OpenFlow SDN (Architecture B)	Centralized Controller (FlowMods)	High visibility, dynamic control	Control plane bottleneck	**Control Traffic:** Very High

9 Conclusion

This paper has demonstrated that HULA achieves significant improvements over ECMP and traditional LVS in both throughput and fairness across all tested topologies. By leveraging the programmable data plane capabilities of modern switches, HULA eliminates the CPU bottlenecks that plague kernel-based solutions and the control plane churn that limits centralized SDN approaches. The architecture successfully decouples state management from packet forwarding, allowing the system to scale to millions of concurrent flows while maintaining deterministic latency.

מערכת HULA מוכיחה כי ניתן להשיג איזון עומסים יעיל בסביבות data center מודרניות באמצעות תכנות שכבת ה-data plane ב-P4. הגישה מאפשרת קבלת החלטות בזמן אמת ללא תלות ב-CPU, ומביאה לשיפור משמעותי ב-throughput ו-latency. בנוסף, היכולת לבצע ריטריביט דינמי בזמן אמת מבלי להפסיק את השירות, מציבה את HULA כפתרון אמין לצרכי תעשייה.

Future work will explore extending HULA to heterogeneous hardware environments and supporting more complex Layer 7 load balancing policies within the data plane. Additionally, integrating security features, such as Web Application Firewall (WAF) logic, directly into the P4 program offers a promising direction for creating high-performance, security-aware load balancers. The results of this study confirm that the future of load balancing lies in the programmable data plane [3].

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